Отчёт по практической работе №3

1. В зонах RIP и OSPF необходимо назначить на интерфейсы маршрутизаторов IP адреса из сетей на /30 (можно воспользоваться шаблоном).

Интерфейсы на роутере C-RIP-1:

```
interface Ethernet0/0
 description "DMikheeva-Main-Router"
 ip address 17.1.1.2 255.255.255.252
!
interface Ethernet0/1
 description "DMikheeva-C-RIP-3"
 ip address 140.116.8.1 255.255.255.252
!
interface Ethernet0/2
 description "DMikheeva-C-RIP-2"
 ip address 140.116.9.1 255.255.255.252
!
interface Ethernet0/3
 description "DMikheeva-Mikrotik"
 ip address 18.1.1.1 255.255.255.252
!
```

Интерфейсы на роутере C-RIP-4:

```
interface Ethernet0/0
 description "DMikheeva-C-RIP-2"
 ip address 213.44.11.2 255.255.255.252
!
interface Ethernet0/1
 description "DMikheeva-C-RIP-3"
 ip address 213.44.10.2 255.255.255.252
!
interface Ethernet0/2
 description "DMikheeva-C-RIP-6"
 ip address 213.44.13.1 255.255.255.252
!
interface Ethernet0/3
 description "DMikheeva-C-RIP-5"
 ip address 213.44.12.1 255.255.255.252
!
```

Интерфейсы на роутере C-RIP-7:

```
interface Ethernet0/0
 description "DMikheeva-C-RIP-6"
 ip address 80.80.80.2 255.255.255.252
!
interface Ethernet0/1
 description "DMikheeva-C-RIP-5"
 ip address 80.80.81.2 255.255.255.252
!
interface Ethernet0/2
 description "DMikheeva-M-RIP-2"
 ip address 80.80.82.1 255.255.255.252
!
interface Ethernet0/3
 description "DMikheeva-M-RIP-1"
 ip address 80.80.84.1 255.255.255.252
!
interface Ethernet1/0
 description "DMikheeva-C-RIP-10"
 ip address 80.80.83.1 255.255.255.252
!
```

Интерфейсы на роутере M-OSPF-10:

```
[admin@DMikheeva-Mikrotik] > ip address pr
Flags: X - disabled, I - invalid, D - dynamic
 #   ADDRESS            NETWORK         INTERFACE
 0   ;;; "DMikheva-Main-Router"
     14.1.1.2/30        14.1.1.0        ether1
 1   ;;; "DMikheva-C-RIP-1"
     18.1.1.2/30        18.1.1.0        ether2
 2   ;;; "DMikheva-M-OSPF-2"
     12.187.10.1/30     12.187.10.0     ether3
 3   ;;; "DMikheva-M-OSPF-3"
     12.208.10.1/30     12.208.10.0     ether4
 4   ;;; "DMikheva-C-OSPF-1"
     12.229.10.1/30     12.229.10.0     ether5
 5   ;;; "DMikheva-C-iBGP-102"
     15.1.1.2/30        15.1.1.0        ether6
 6   10.255.255.1/32    10.255.255.1    loopback
```

Интерфейсы на роутере M-OSPF-3:

```
[admin@DMikheeva-M-OSPF-3] > ip address pr
Flags: X - disabled, I - invalid, D - dynamic
 #   ADDRESS            NETWORK         INTERFACE
 0   ;;; DMikheeva-Mikrotik
     12.208.10.2/30     12.208.10.0     ether1
 1   ;;; DMikheeva-M-OSPF-2
     12.208.11.2/30     12.208.11.0     ether2
 2   ;;; DMikheeva-M-OSPF-4
     12.208.12.1/30     12.208.12.0     ether3
 3   ;;; DMikheeva-C-OSPF-2
     12.208.13.1/30     12.208.13.0     ether4
 4   10.255.255.3/32    10.255.255.3    loopback
```

Интерфейсы на роутере M-OSPF-5:

```
[admin@DMikheeva-M-OSPF-5] > ip address pr
Flags: X - disabled, I - invalid, D - dynamic
 #   ADDRESS            NETWORK         INTERFACE
 0   ;;; DMikheeva-M-OSPF-4
     12.208.15.2/30     12.208.15.0     ether1
 1   ;;; DMikheeva-C-OSPF-3
     12.208.17.1/30     12.208.17.0     ether2
 2   ;;; DMikheeva-C-OSPF-5
     12.208.16.1/30     12.208.16.0     ether3
 3   ;;; DMikheeva-C-OSPF-2
     12.208.14.2/30     12.208.14.0     ether4
 4   10.255.255.5/32    10.255.255.5    loopback
```

2. В зоне RIP - настроить работу протокола динамической маршрутизации RIP.

Трассировка от роутера C-RIP-1 до роутера C-RIP-10:

```
DMikheeva-C-RIP-1#traceroute 80.80.83.2
Type escape sequence to abort.
Tracing the route to 80.80.83.2
VRF info: (vrf in name/id, vrf out name/id)
  1 140.116.8.2 2 msec
    140.116.9.2 1 msec
    140.116.8.2 1 msec
  2 93.160.249.2 3 msec
    75.24.43.2 6 msec
    93.160.249.2 3 msec
  3 80.80.80.2 3 msec
    80.80.81.2 5 msec
    80.80.80.2 2 msec
  4 80.80.83.2 5 msec 5 msec *
```

Трассировка от роутера C-RIP-10 до роутера C-RIP-4:

```
DMikheeva-C-RIP-10#traceroute 213.44.12.1
Type escape sequence to abort.
Tracing the route to 213.44.12.1
VRF info: (vrf in name/id, vrf out name/id)
  1 80.80.83.1 1 msec
    80.80.82.2 4 msec
    80.80.83.1 1 msec
  2 93.160.240.1 2 msec
    80.80.81.1 4 msec
    93.160.240.1 3 msec
  3 213.44.12.1 9 msec 3 msec *
```

3. В зоне OSPF - настроить работу протокола динамической маршрутизации OSPF.

Трассировка от роутера M-OSPF-10 до роутера C-OSPF-5:

```
[admin@DMikheeva-Mikrotik] > tool traceroute 12.229.10.18
 # ADDRESS                          LOSS SENT    LAST     AVG    BEST   WORST
 1 12.187.10.2                        0%   11   3.4ms     2.8     1.7     7.1
 2 12.187.11.2                        0%   11   4.3ms       5     2.5    10.2
 3 12.229.10.18                       0%   11   4.6ms     4.7       3      10
```

Трассировка от роутера C-OSPF-5 до роутера C-OSPF-1:

```
DMikheeva-C-OSPF-5#traceroute 12.229.10.5
Type escape sequence to abort.
Tracing the route to 12.229.10.5
VRF info: (vrf in name/id, vrf out name/id)
  1 12.208.16.1 1 msec 2 msec 1 msec
  2 12.208.14.1 2 msec 9 msec 2 msec
  3 12.229.10.5 3 msec *  9 msec
```

4. Связать зоны RIP, OSPF и Static.

Трассировка от роутера C-RIP-10 до роутера C-OSPF-5:

```
DMikheeva-C-RIP-10#traceroute 12.187.12.2
Type escape sequence to abort.
Tracing the route to 12.187.12.2
VRF info: (vrf in name/id, vrf out name/id)
  1 80.80.83.1 2 msec
    80.80.82.2 2 msec
    186.229.48.2 3 msec
  2 80.80.81.1 3 msec
    93.160.240.1 2 msec
    75.24.44.1 3 msec
  3 93.160.249.1 2 msec 5 msec
    75.24.43.1 6 msec
  4 140.116.9.1 2 msec 10 msec
    140.116.8.1 5 msec
  5 18.1.1.2 17 msec 10 msec 10 msec
  6 12.208.10.2 9 msec 10 msec 9 msec
  7 12.208.12.2 7 msec 34 msec 13 msec
  8 12.187.12.2 17 msec 10 msec *
```

Трассировка от роутера C-OSPF-4 до роутера C-RIP-10:

```
DMikheeva-C-OSPF-4#trace 80.80.83.2
Type escape sequence to abort.
Tracing the route to 80.80.83.2
VRF info: (vrf in name/id, vrf out name/id)
  1 12.229.10.13 1 msec
    12.229.10.18 2 msec
    12.229.10.13 1 msec
  2 12.208.15.1 19 msec
    12.229.10.9 2 msec
    12.208.15.1 3 msec
  3 12.229.10.5 7 msec
    12.208.11.1 8 msec
    12.229.10.5 3 msec
  4 12.208.10.1 8 msec 7 msec 10 msec
  5 18.1.1.1 10 msec 11 msec 5 msec
  6 140.116.9.2 6 msec 9 msec 10 msec
  7 93.160.249.2 11 msec 12 msec 5 msec
  8 80.80.81.2 8 msec 18 msec 8 msec
  9 80.80.83.2 8 msec 19 msec *
```

Трассировка от роутера C-RIP-7 до роутера C-R-3-3:

```
DMikheeva-C-RIP-7#trace 169.31.3.2
Type escape sequence to abort.
Tracing the route to 169.31.3.2
VRF info: (vrf in name/id, vrf out name/id)
  1 80.80.81.1 2 msec
    80.80.80.1 2 msec
    80.80.81.1 3 msec
  2 75.24.43.1 4 msec
    93.160.249.1 3 msec
    75.24.43.1 2 msec
  3 140.116.9.1 3 msec
    140.116.8.1 3 msec
    140.116.9.1 5 msec
  4 17.1.1.1 3 msec 17 msec 2 msec
  5 10.38.0.1 6 msec 6 msec 4 msec
  6 16.80.3.1 4 msec 11 msec *
```

Трассировка от роутера C-RIP-6 до компьютера PC34:

```
DMikheeva-C-RIP-6#trace 10.210.20.10
Type escape sequence to abort.
Tracing the route to 10.210.20.10
VRF info: (vrf in name/id, vrf out name/id)
  1 75.24.43.1 1 msec 2 msec 1 msec
  2 140.116.8.1 10 msec 4 msec 3 msec
  3 17.1.1.1 8 msec 3 msec 16 msec
  4 10.210.20.10 18 msec 8 msec 9 msec
```

Трассировка от роутера C-OSPF-4 до роутера C-R-4-1:

```
DMikheeva-C-OSPF-4#trace 10.187.11.2
Type escape sequence to abort.
Tracing the route to 10.187.11.2
VRF info: (vrf in name/id, vrf out name/id)
  1 12.229.10.18 2 msec
    12.229.10.13 1 msec
    12.229.10.18 4 msec
  2 12.229.10.9 3 msec
    12.208.15.1 7 msec
    12.229.10.9 2 msec
  3 12.208.11.1 8 msec
    12.229.10.5 3 msec
    12.208.11.1 11 msec
  4 12.229.10.1 7 msec 6 msec 14 msec
  5 14.1.1.1 9 msec 5 msec 7 msec
  6 10.122.0.1 6 msec 18 msec 7 msec
  7 10.166.10.1 18 msec 24 msec *
```

Трассировка от роутера M-OSPF-3 до компьютера PC24:

```
[admin@DMikheeva-M-OSPF-3] > tool trace 10.210.40.10
 # ADDRESS                          LOSS SENT    LAST     AVG    BEST   WORST
 1 12.208.10.1                        0%    8   2.3ms     2.9     1.4     4.8
 2 14.1.1.1                           0%    8   6.5ms     4.3     2.6     9.5
 3 10.210.40.10                       0%    8   4.4ms     6.5     2.8    14.4
```